

Mustafa Abdallah

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Experience

Software Engineer, [Upwork](#)

Jan 2021 – Oct 2023

- **Built** custom software solutions for **multiple international clients**, from **Ireland, Kenya, the UK, the USA, Canada, and Germany**, where I was responsible for the entire software development lifecycle, from the initial discussions with clients to understand their requirements, to development, deployment, and maintenance.
- **Refactored** Django dashboard, as measured by improved feature support, by migrating the backend to **Python, Django, and PostgreSQL**, and frontend from **Django templates, HTML, and CSS** to **TypeScript, React.js, and Redux**
- **Modernized** an Android application and improved app maintainability, as measured by migration of a legacy Android app from **Java and XML** to **Kotlin, Jetpack Compose, and MVVM architecture**, by refactoring a legacy **Java** codebase
- **Delivered** new backend features in a legacy system, as measured by seamless integration with existing database logic, by analyzing client requirements and implementing modular APIs using **Python, Flask, MySQL, and SQLAlchemy**
- **Automated** responses to common user queries, as measured by successful handling of routine user requests, by building a **WhatsApp chatbot** integrated with a **Flask backend API** using **Node.js, TypeScript, Twilio, and Dialogflow**
- **Improved** login flexibility, as measured by support for multiple authentication methods (email/password, PIN, biometric fingerprint), by implementing a multi-layer authentication system in a **Flutter** app with a **Flask** backend
- **Enabled** access control across user roles, as measured by support for custom roles with defined permissions, by implementing a role-based access control (RBAC) system in a **Flutter** mobile app with a **Python** and **Flask** backend
- **Migrated** DevOps and cloud deployment for a **Django** backend from **Heroku** to **DigitalOcean**, as measured by automated multi-environment deployments (dev, staging, prod) and streamlined infrastructure management, by creating infrastructure with **Terraform**, containerizing with **Docker**, and implementing **CI/CD pipelines** using **CircleCI**
- **Implemented** a cloud infrastructure with DevOps on **AWS (Amazon Web Services)** for **Django REST APIs**, as measured by fully automated deployments, by building **Terraform**-managed infrastructure (EC2, RDS, S3, ECR), containerizing services with **Docker/Docker Compose**, and implementing **CI/CD pipelines** with **CircleCI**
- **Created** a cloud infrastructure with DevOps on **Microsoft Azure** for **Flask REST APIs**, as measured by fully automated deployments, by designing **Terraform**-managed resources (Virtual Machines, VNet, PostgreSQL, Blob Storage, ACR), containerizing services with **Docker**, and implementing **CI/CD pipelines** with **GitHub Actions**

Founder, [Tarwiiga](#)

Jan 2024 – Present

- **Helped** local service providers grow their customer base, as measured by generating high-quality inbound leads, by **building custom business websites** with **no-code platforms**, and launching targeted Google Ads campaigns

Projects

AI-Powered Campaign Builder | [Demo](#)

May 2024 – Oct 2024

- **Scaled** Google Ads campaign creation using **Generative AI**, as measured by reducing manual campaign setup from 3 days to under 1 minute, by developing a platform using **TypeScript/Next.js, Python/FastAPI, LLM, LangChain, OpenAI GPT, and Google Gemini API**, deployed as a **serverless** architecture on **Google Cloud Run** with **Docker**

AI Conversational Assistant | [Demo](#)

May 2024 – Oct 2024

- **Enabled** expert-level Google Ads question answering, as measured by retrieval-based responses over a **5K bilingual (Arabic/English) knowledge base**, by developing a **Retrieval-Augmented Generation (RAG)** pipeline using **Python, LangChain, FastAPI, Chroma vector database, Hugging Face embeddings, and LLMs (OpenAI GPT, Google Gemini)**

Education

Bachelor of Science, Computers and Systems Engineering – Minia University

2015 – 2020